

SPI MASTER CONTROLLER

RTL-to-GDSII Implementation

A Complete Digital VLSI Physical Design Project

Project	SPI-Controller-RTL-to-GDSII
Design Type	Sequential / Clocked RTL Design
Technology	Sky130
Implementation Flow	OpenLane v1.0.2
Top Module	spi_master
Protocol	8-bit SPI, Mode 0 (CPOL=0, CPHA=0)
Physical Design	Synthesis → Floorplan → Placement → CTS/Clock Analysis → Routing → Signoff
Final Deliverable	GDSII layout with signoff reports

Final Project Report

Prepared for the completed SPI Master Controller implementation and its recorded design, simulation and signoff results.

1. Abstract

This project implements an SPI Master Controller as a sequential digital design and carries it through a complete RTL-to-GDSII physical design flow using the Sky130 technology and OpenLane v1.0.2. The controller is a clocked design so that RTL functionality, physical implementation and clock-aware timing analysis can be demonstrated.

The completed flow includes RTL development, simulation, waveform verification, logic synthesis, floorplanning, I/O placement, power planning, placement, clock-related implementation/analysis, routing, static timing analysis, physical signoff and final GDSII generation.

The SPI controller implements an 8-bit transaction using SPI Mode 0 (CPOL=0, CPHA=0). The design generates SCLK and CS, transmits serial data through MOSI and samples received data through MISO.

2. Project Objectives

- Design and verify a practical sequential SPI Master Controller at RTL.
- Implement an 8-bit SPI transaction using SPI Mode 0.
- Demonstrate a clocked VLSI design with clock-aware physical implementation and timing analysis.
- Run the design through an end-to-end RTL-to-GDSII flow using OpenLane.
- Generate a physical layout using the Sky130 standard-cell technology.
- Perform detailed static timing analysis including multiple operating corners.
- Complete physical signoff checks and generate a final GDSII database.
- Organize implementation results, reports and evidence as a GitHub portfolio project.

3. Design Overview

The design is an SPI Master Controller. SPI is a synchronous serial communication protocol in which a master controls the serial clock and chip-select signal while transmitting and receiving serial data.

3.1 Main Design Characteristics

Item	Project implementation
Design	SPI Master Controller
Top-level module	spi_master
Design class	Sequential / clocked
Clock	System clock used for controller sequencing and SPI clock generation
Protocol	SPI Mode 0 (CPOL=0, CPHA=0)
Data width	8-bit transaction
Serial transmit	MOSI output
Serial receive	MISO input
Clock output	SCLK
Chip select	CS
Control/status	Busy and done indication
Physical implementation	Standard-cell based Sky130 flow

3.2 Functional Operation

- The controller receives 8-bit transmit data at RTL.
- A system clock drives the sequential SPI transaction process.
- The internal clock-divider logic generates the SPI serial clock.
- The MOSI output changes on the falling edge of SCLK.
- The MISO input is sampled on the rising edge of SCLK.
- The bit counter and shift registers sequence the 8-bit transfer.
- The controller asserts the chip-select signal during the transaction.
- After the final bit, the controller returns to the idle state and asserts the done indication.

3.3 RTL-to-GDSII Flow

The implementation follows the standard digital physical-design sequence: RTL → synthesis → floorplan → I/O placement → power planning → placement → clock-related implementation/analysis → routing → extraction/STA → signoff → GDSII.

4. RTL Verification and Simulation

The SPI Master Controller RTL was verified before physical implementation. A dedicated Verilog testbench was used to exercise an 8-bit SPI transaction and generate a waveform dump. The simulation result was verified using Icarus Verilog and the waveform was inspected using GTKWave.

Verification Evidence

Evidence	Project file/evidence
SPI RTL	spi_master RTL source
Testbench	tb_spi_master testbench
Simulation	SPI simulation result
Waveform	GTKWave waveform evidence
Physical layout	Final GDS/KLayout layout evidence

Recorded simulation result

TX DATA = 10110010 RX DATA = 11001101 BUSY = 0 DONE = 1

TEST RESULT = PASS

The simulation stage establishes functional confidence at RTL before the design is passed to synthesis and physical implementation. The waveform screenshot is retained as project evidence together with the RTL and simulation screenshots.

5. Physical Design Implementation

5.1 Synthesis

The SPI RTL was synthesized into a gate-level standard-cell representation using the OpenLane flow and Sky130 standard-cell libraries. The synthesis stage generated area/timing-related reports and the gate-level representation required for subsequent physical design.

5.2 Floorplanning and I/O Placement

The design was assigned a physical core/die region and the I/O interface was placed for the subsequent power, placement and routing stages.

5.3 Power Planning and Placement

Power distribution structures were generated and the synthesized standard cells were placed within the floorplan. The placement results were preserved in the project repository as implementation evidence.

5.4 Clock-Related Implementation

Unlike a combinational design, this SPI controller is clocked. Therefore clock-aware analysis and CTS-related implementation are applicable to the design.

5.5 Routing

The placed design was routed to create physical interconnects between standard cells and I/O pins. The completed run generated routed DEF and ODB databases together with extracted and signoff artifacts.

6. Physical Design Results

The completed SPI implementation produced the physical-design and signoff evidence required for the project. The exact numerical area, power and timing values are retained in the generated OpenLane reports and detailed STA material.

Metric	Recorded result
Standard-cell count	See synthesis report
Total cell area	See synthesis report
Total power	See power report
Internal power	See power report
Switching power	See power report
Leakage power	See power report
TNS	See STA report
WNS	See STA report
Worst setup slack	See STA report
Worst hold slack	See STA report

The detailed reports are preserved in the project repository rather than replacing recorded implementation data with estimated values.

7. Static Timing Analysis (STA)

Static timing analysis was completed for the clocked SPI Master Controller. The project preserves detailed STA material rather than only a single timing number. The available timing evidence includes separate maximum, minimum and nominal operating-corner reports together with a combined all-corner report.

7.1 STA Report Organization

Report	Purpose
SS corner reports	Slow-slow operating-corner timing analysis
TT corner reports	Typical-typical operating-corner timing analysis
FF corner reports	Fast-fast operating-corner timing analysis
Detailed STA reports	Checks, maximum timing, minimum timing, power, skew and summary analysis
Combined all-corner STA	Combined detailed multi-corner STA evidence

7.2 Timing Summary

Timing item	Final recorded result
TNS	Available in STA report
WNS	Available in STA report
Worst setup slack	Available in STA report
Worst hold slack	Available in STA report
STA coverage	Detailed SS/TT/FF and combined all-corner reports retained

The detailed all-corner STA material is retained as generated project evidence. It documents that multi-corner timing analysis was performed and that the detailed timing artifacts are part of the final project.

8. Physical Signoff

The final implementation passed the recorded physical signoff checks. The signoff evidence is stored in the project results directory.

Signoff check	Recorded result
DRC	0 violations
LVS	0 errors
XOR / layout comparison	0 differences
Antenna	Antenna verification report generated
Final GDSII	Generated successfully
KLayout GDS	Generated successfully

8.1 Final Generated Artifacts

- spi_master.gds — final GDSII database
- spi_master.klayout.gds — KLayout-compatible GDS
- spi_master.lef — layout exchange format
- spi_master.lib — timing/power library representation
- spi_master.sdf — extracted timing information
- spi_master.spice — extracted SPICE representation
- spi_master.mag — Magic layout representation
- spi_master.rpt — final implementation report

9. Project Repository Organization

The SPI Master Controller project is maintained as a separate GitHub repository so that it can be presented independently from the previously completed UART and 4-bit ALU projects.

Directory / item	Purpose
rtl	SPI Master Controller source
simulation	Testbench and simulation material
openlane	OpenLane configuration and flow material
results/floorplan	Floorplan-related reports/artifacts
results/placement	Placement DEF/ODB and reports
results/routing	Routing DEF/ODB and routing reports
results/sta	Detailed STA and all-corner timing reports
results/signoff	DRC/LVS/XOR/antenna and final signoff reports
results/final	Final GDSII and extracted deliverables
screenshots	RTL, simulation, waveform and layout evidence
docs	Final project documentation

The repository contains reproducible implementation artifacts and presentation-oriented evidence for the complete SPI RTL-to-GDSII project.

10. Technical Learning and Outcomes

- Implemented a sequential SPI Master Controller at RTL.
- Understood SPI communication and Mode 0 timing with CPOL=0 and CPHA=0.
- Implemented 8-bit serial transmission and reception.
- Implemented SPI clock generation using clock-divider logic.
- Used transmit and receive shift registers and a bit counter for serial data sequencing.
- Worked through the major OpenLane physical-design stages.
- Handled floorplanning, placement, routing and clock-related implementation for a sequential design.
- Generated and organized multi-corner STA reports.
- Performed physical signoff checks including DRC, LVS and XOR comparison.
- Generated industry-standard physical-design deliverables such as GDSII, LEF, LIB, SDF and SPICE.
- Organized the complete project as a GitHub portfolio repository.

11. Conclusion

The SPI Master Controller project successfully demonstrates an end-to-end digital VLSI implementation flow from RTL through physical signoff and GDSII generation. The project is stronger than a purely combinational example because it includes a clocked sequential communication controller and corresponding timing analysis.

The completed implementation includes functional verification, physical design databases, detailed multi-corner STA material and signoff evidence. The RTL simulation achieved a recorded PASS result for the 8-bit SPI transaction, with TX DATA = 10110010, RX DATA = 11001101, BUSY = 0 and DONE = 1.

The final physical implementation generated the required layout and signoff outputs, including GDSII, and the recorded signoff checks achieved zero DRC violations, zero LVS errors and zero XOR differences.

The final result is a complete portfolio-ready RTL-to-GDSII project demonstrating practical exposure to SPI RTL design, simulation, synthesis, physical implementation, timing analysis and physical verification using Sky130 and OpenLane.